

49%

INCREASE IN TENDERED CAPACITY OF HYBRID PROJECTS (RENEWABLE WITH STORAGE) IN 2024, FROM 12% IN 2021

last two years. In comparison, energy storage system costs have dipped 40%, with a record drop in the price of lithium-ion batteries, the dominant storage medium. Tendered capacity of hybrid projects (renewable with storage) increased to over 49% in 2024 from 12% in 2021.

Does this mean there are no hurdles in India's march to become a renewable energy powerhouse? Praveer Sinha, Chairman and Managing Director of Tata Power Company Ltd, says despite falling prices and government incentives, the growing pipeline of under-construction projects is raising concern that developers are quoting artificially low prices.

"Aggressive underbidding not only skews the auction but also places successful commissioning of projects in jeopardy. This is further complicated by delays in PPA signing and grid connections...," Sinha told *Business Today*.

True, energy storage has many powerful arguments in its favour. India is blessed with plentiful sunlight during the day for generating solar power. Yet, meeting peak demand in afternoons and early evenings remains a challenge. A storage system reduces the cost of running the power grid and helps businesses avoid expensive disruptions due to power outages. Household consumers can use stored energy, especially that generated from low-cost renewable sources such as solar during the day, apart from wind, to reduce expenses on storing perishable foods and medicines. Battery storage systems can ensure round-the-clock availability of green energy at competitive tariffs. Projects are offering solar energy with six hours of peak power storage.

Financing Complexities

Even so, concerns loom over the booming storage market, which requires an investment of ₹3.68 lakh crore between 2027 and 2032. The primary challenge is high capital costs and financing complexities of BESS projects despite the recent reductions in battery pack prices to around \$55 per kilo Watt hour or kWh. Also, no big project has been commissioned that can lift confidence in imported China-made batteries for large-scale BESS projects in the hot and humid geographies of Gujarat, Tamil Nadu and Rajasthan.

Experts say standardised procurement frameworks, transparent tariff structures and well-defined dispatch guidelines are essential to provide developers the confidence to invest.

The government has released

an advisory that all new solar projects will need to come with two hours of storage, sending a clear signal that the sector will keep growing. It has also announced two rounds of VGF—first with 30% capex and second with 16% capex and waiver of inter-state transmission charges for BESS projects (co-located with renewable energy plants) commissioned on or before June 30, 2028.

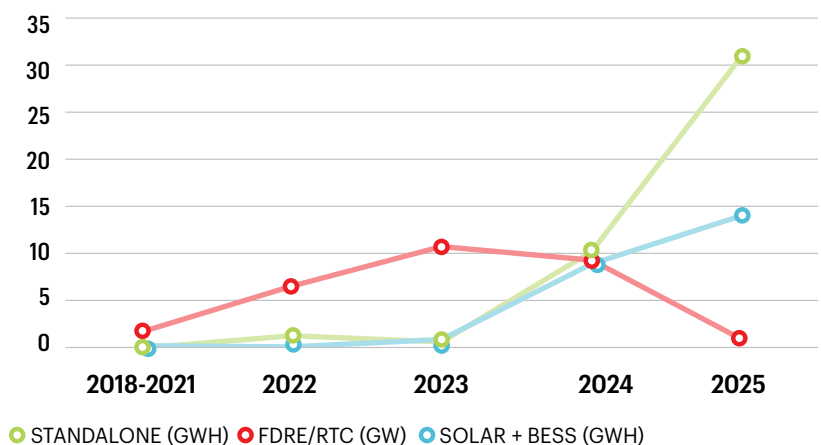
Tumbling Prices

BESS prices fell 22.3% from a 20 Mega Watt (MW) auction in New Delhi overseen by The Energy and Research Institute in September

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CAPACITY STORAGE

- Standalone BESS is the top type of tender according to capacity



NOTE GWH IS GIGA-WATT-HOUR; FDRE/RTC IS FIRM AND DISPATCHABLE RENEWABLE ENERGY/ROUND THE CLOCK **SOURCE** TENDER NODAL AGENCIES